

Exercises in logical proposition:

1) Premises: $p \rightarrow q$
 p

Prove: q

Solution:

Premises:

1. $p \rightarrow q$

2. p

reasoning:

From premise (1) and premise (2) by Modus Ponens
we conclude q

2) Premises: $p \rightarrow q$
 $\neg q$

Prove: $\neg p$

Solution:

Premises:

1. $p \rightarrow q$

2. $\neg q$

reasoning:

From premise (1) and (2) by Modus Tollens rule
we conclude $\neg p$

3) Premises: $P \vee Q$
 $\neg P$

Prove: Q

Solution:

Premises:

1. $P \vee Q$

2. $\neg P$

reasoning: From premise (1) and (2) by disjunctive
Syllogism we conclude Q .

4) Premises: $P \rightarrow Q$
 $Q \rightarrow R$
 P

Prove: R

Solution:

Premises:

1. $P \rightarrow Q$

2. $Q \rightarrow R$

3. P

reasoning: From premise (3) and (1) by Modus Ponens rule
we conclude $Q \Rightarrow$ From Q and premise (2)
by Modus Ponens rule we get R

5) Premises: $P \rightarrow Q$
 $Q \rightarrow R$
 $\neg R$

Prove: $\neg P$

Solution:

Premises:

1. $P \rightarrow Q$

2. $Q \rightarrow R$

3. $\neg R$

reasoning: 1. From premise (3) and (2) by Modus Tollens rule we conclude $\neg Q$.

2. From premise (1) and $\neg Q$ by Modus Tollens rule we conclude $\neg P$

6) Premises: $P \vee Q$
 $Q \rightarrow R$
 $\neg P$

Prove: R

Solution:

Premises:

1. $P \vee Q$

2. $Q \rightarrow R$

3. $\neg P$

reasoning:

1. From premise (3) and (1) by disjunctive syllogism
we get Q

2. From premise (2) and Q by Modus Ponens
we get R

7) Premises: $P \rightarrow Q$
 $R \rightarrow P$
 R

Prove: Q

Solution:

Premises:

1. $P \rightarrow Q$

2. $R \rightarrow P$

3. R

reasoning:

1. From premiss (3) and (2) by Modus Ponens
we get P

2. From premise (1) and P by Modus Ponens
we get Q

8) Premises: $P \rightarrow Q$
 $Q \rightarrow R$
 $R \rightarrow S$
 P

Prove: S

Solution:

Premises:

1. $P \rightarrow Q$
2. $Q \rightarrow R$
3. $R \rightarrow S$
4. P

reasoning

1. From premiss (1) and (4) by Modus Ponens rule we get Q
2. From premiss (2) and Q by Modus Ponens rule we get R
3. From premiss (3) and R by Modus Ponens rule we get S

$$9) \quad p \vee Q$$

$$p \rightarrow R$$

$$Q \rightarrow R$$

Prove R

Solution:

Premises:

1. $p \vee Q$
2. $p \rightarrow R$
3. $Q \rightarrow R$

reasoning:

1. From premise (1) we have two cases p or Q true or both, but if one of them true all the proposition is true
2. if p is true (case 1) $\Rightarrow p \vee Q = \text{true}$ and
by Modus Ponens from Premis (2) we get R
3. if Q is true (case 2) so from Premis (3) by
by Modus ponens we get R
4. we get R in the Both cases by disjunction
elimination we conclude R

10) $P \rightarrow Q$
 $Q \rightarrow R$
 $S \rightarrow P$
 $\neg R$

Prove: $\neg S$

Solution:

Premises:

1. $P \rightarrow Q$
2. $Q \rightarrow R$
3. $S \rightarrow P$
4. $\neg R$

reasoning:

1. From premise (2), by Modus Tollens we get $\neg Q$
2. From premise (1) ~~P~~ and $\neg Q$, by Modus Tollens we get $\neg P$
3. From premise (3) and $\neg P$, by Modus Tollens we get $\neg S$
4. we conclude $\neg S$

Proved ✓

